

Project Planning & Management

Project proposal:

analyze loan risk and customer behavior related to loan acquisition, including the reasons behind taking loans and the factors influencing customers' decisions. Using Power BI, the project extracts meaningful insights to help the bank to identify high-risk or unreliable customers and make better decisions regarding loan approvals and credit risk management.

This helps improve financial decision-making and reduce potential credit losses.

Task Assignment & Rules

The distribution of team member responsibilities was defined as following:

- 1- Mohamed Hany & Mrawan Said
 - Responsible for data cleaning and Preprocessing.
- 2- Malak Elsayed & Shahd Fagelnour
 - Responsible for data modeling.
- 3- Sandra Bassem & Login tamer
 - Responsible for data visualization.

Project plan:

The project will be completed through the following steps:

1) Data Collection

The dataset used in this project is obtained from Kaggle:

<https://www.kaggle.com/datasets/zaurbegiev/my-dataset>

2) Data Cleaning & Preprocessing

During this step, the dataset was cleaned to ensure accuracy and consistency. The following tasks was performed:

- Promoted headers
- Standardized column names and formats
- Checked and corrected data types
- Extract text
- Replaced values
- Filtered irrelevant data
- Handle missing values

✓ This step ensured that the data was clean, reliable and ready for analysis and modeling.

4) Data Modeling

In this step, the data model was designed to organize and structure the dataset for efficient analysis in Power BI.

- The model follows a Star Schema structure
- The main tables used are:
 - fCredit (Fact table)
 - dCustomer (Dimension table)
- A one-to-many relationship was created (Single direction) between fCredit and dCustomers(one customer can have many loans)
- A separate Measures table was created to store all the following DAX calculations:

1- Average Annual Income

- The average yearly income of all borrowers.
- Used to understand customers' earning levels.

2- Average Credit Balance

- The average of current credit balance.
- Shows how much debt customers still owe.

3- Average Credit Score

- The average credit score of borrowers.
- Indicates overall creditworthiness and risk level.

4- Average Debt-to-income Ratio

- The average ratio of debt compared to income.
- Helps measure customers' ability to handle debt.

5- Average Loan Amount

- The average of current loans amount.
- Shows typical loan size in the dataset.

6- Average monthly debt

- The average monthly debt of borrowers.
- Indicates financial burden on customers.

7- Average year in current job

- The average number of years borrowers stayed in their job.
- Reflects job stability.

8- Charged off Loans

- Number of loans that were not repaid (defaulted).
- Represents total bad loans.

9- Charged off Ratio

- Percentage of loans that were charged off ($\text{charged off} / \text{total loans}$).
- Shows the proportion of risky loans.

10- Default Rate

- Percentage of defaulted loans out of total loans.
- Used to measure overall level of loans defaulted.

11- Fully Paid Loans

- Number of loans that were completely repaid(count fully paid loans)
- Indicates successful repayments.

12- Fully Paid Ratio

- Percentage of loans fully paid ($\text{fully paid} / \text{total loans}$).
- Shows how many loans ended successfully.

13- Num of Borrowers

- Total number of customers in the dataset(Distinctcount customer ID)

- Used as a base for analysis.

14- Total Loan Amount

- Sum of all loan amounts issued.
- Represents total lending volume.

- Some calculated columns were created to enrich the analysis and were hidden to improve model clarity and performance:

1. Annual Income Availability

- Indicates whether income data is available or missing.

2. Annual Income Category

- Classifies customers into Low, Average, and High income groups.

3. Credit Score Availability

- Indicates whether credit score data is available.

4. Credit Score Category

- Classifies credit scores into Fair, Good, Very Good, and Needs Clarification.

5. Adjusted Current Loan Amount

- Replaces abnormal values (e.g., 99999999) with a more realistic median value.

6. Debt-to-Income Category

- Classifies customers into Low, Medium, and High risk.

7. Debt-to-Income Ratio

- Calculated as $(\text{Monthly Debt} \times 12) / \text{Annual Income}$.

8. Loan Amount Category

- Groups loans into Small, Average, and Large amounts.

9. Months Since Last Delinquent Category

- Classifies delinquency status into No Delinquency, Recent, and Old.

10. Years-in-Job Category

-Categorizes employment experience into New Employee, Low, Medium, and High Experience.

- A screenshot of some calculated columns

The screenshot shows a data table interface with a 'Table tools' ribbon and a 'Column tools' ribbon. The table has columns for 'Loan Amount Category', 'Years-in-Job Category', 'Debt-To-Income Ratio', 'Debt-to-Income Category', 'Month Since Last Delinquent Category', 'Annual Income availability', and 'credit score availability'. The 'credit score availability' column is calculated using the formula: `= IF(ISBLANK('credit_train csv'[Credit Score]),"Not Available","Available")`. The table contains 81,823 rows, with 64,831 filtered rows. The 'credit score availability' column shows 'Available' for all rows.

Loan Amount Category	Years-in-Job Category	Debt-To-Income Ratio	Debt-to-Income Category	Month Since Last Delinquent Category	Annual Income availability	credit score availability
Large Amount	Low Experience	5.9%	Low Risk	No Delinquency	Available	Available
Large Amount	Low Experience	28.8%	Medium Risk	No Delinquency	Available	Available
Large Amount	Medium Experience	20.7%	Medium Risk	No Delinquency	Available	Available
Large Amount	New Employee	29.7%	Medium Risk	No Delinquency	Available	Available
Large Amount	Low Experience	11.7%	Low Risk	No Delinquency	Available	Available
Large Amount	Low Experience	22.1%	Medium Risk	No Delinquency	Available	Available
Large Amount	Low Experience	18.3%	Low Risk	No Delinquency	Available	Available
Large Amount	High Experience	16.1%	Low Risk	No Delinquency	Available	Available
Large Amount	Low Experience	27.2%	Medium Risk	No Delinquency	Available	Available
Large Amount	High Experience	22.1%	Medium Risk	No Delinquency	Available	Available
Large Amount	Medium Experience	8.5%	Low Risk	No Delinquency	Available	Available
Large Amount	High Experience	15.2%	Low Risk	No Delinquency	Available	Available
Large Amount	New Employee	10.8%	Low Risk	No Delinquency	Available	Available
Large Amount	New Employee	12.5%	Low Risk	No Delinquency	Available	Available
Large Amount	New Employee	5.5%	Low Risk	No Delinquency	Available	Available
Large Amount	High Experience	8.8%	Low Risk	No Delinquency	Available	Available
Large Amount	Low Experience	16.9%	Low Risk	No Delinquency	Available	Available
Large Amount	Medium Experience	12.6%	Low Risk	No Delinquency	Available	Available
Large Amount	Medium Experience	23.4%	Medium Risk	No Delinquency	Available	Available
Large Amount	Low Experience	20.1%	Medium Risk	No Delinquency	Available	Available

✓ These transformations and calculated fields were created to improve data quality, handle missing and inconsistent values, and enable more accurate analysis and reporting.

Risk Assessment & Mitigation plan

1 – risks in cleaning data:

- Missing values in key data (income, age, loan amount).
- Duplicate records.
- Inconsistent data types.

Mitigation:

- Handle missing values by removing or establishing categories for them in the calculated columns.
- Remove duplicate records.
- Standardize data types.

2- Risks in data modeling

- Table division

Mitigation:

We Take a duplicate from the original table and removing the unnecessary columns to generate the customer table to improve data structure and model performance.

Requirement Gathering

Stakeholder Analysis

The main beneficiaries from this project idea:

1- Lenders (such as Banks – financial institutions)

- Are the main beneficiary.
- They use the model to decide whether to approve or reject loan applications.
- This will help them to assess customer risk, reduce the loan default and make better loan decisions.

2- Investors

- They are indirect beneficiary
- They use the results to evaluate the risk level of the loan portfolio.
- They analyze the performance and stability of loans before making investment decisions.